

AI-Powered Decision Support for Climate-Resilient Agriculture in Sri Lanka

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ABSTRACT

Climate variability driven by the El Niño-Southern Oscillation (ENSO) increasingly threatens Sri Lanka's interconnected water, energy, and food systems, with smallholder farmers bearing the heaviest impact due to limited access to timely climate and agronomic information. This research develops, implements, and evaluates an integrated machine-learning decision-support system combining ENSO-informed rainfall forecasting using LSTM networks, irrigation-scheduling optimization using Random Forest and Gradient Boosting ensembles, satellite-based crop-yield prediction using convolutional neural networks, and a farmer-facing advisory layer delivered through a mobile application, SMS alerts, and community radio. The system was piloted across five agroecological zones with three hundred participating farmers over two growing seasons, benchmarked against three hundred control farmers. Results show that three-month rainfall forecasts achieved sixty-eight to seventy-two percent directional accuracy, irrigation recommendations reduced water use by an average of twenty-seven percent while maintaining or improving yields, and crop-yield models explained up to eighty-two percent of yield variance. Field outcomes included thirteen to twenty percent higher crop yields, a thirty-four percent increase in net farm income, and a twenty-seven-percentage-point improvement in production retention during the 2023-2024 El Niño drought relative to non-users. Extension-worker endorsement, visible early results, farmer networks, and local customization emerged as the strongest drivers of sustained adoption. These findings indicate that AI-based decision support can meaningfully strengthen climate resilience for South Asian smallholder agriculture, and the paper concludes with institutional and policy recommendations for national-scale deployment.

CHAPTER 1: INTRODUCTION

1.1 Background and Context

Farmers around the world are feeling the effects of unpredictable weather, but the situation is especially tough for those in developing countries with tropical climates. In Sri Lanka, agriculture is not just a part of the economy; it is the main source of income for millions of people living in rural areas, and about 18% of the national workforce relies on it for their jobs. The country's vulnerability stems from its geographic location and heavy dependence on monsoon rains, which have become increasingly unpredictable in recent years [1].

The El Niño–Southern Oscillation (ENSO) counts as a big climatic driver. It starts from Pacific Ocean temperature shifts working together with changes in atmospheric circulation. When ENSO kicks in, precipitation patterns get altered in measurable ways across wide areas, including South Asia. During El Niño phases, Sri Lanka often faces a notable drop in rainfall; sometimes it can fall around thirty to fifty percent under the longer-term averages, especially during the periods that matter most for farming seasons [2]. These rainfall shortfalls don't stay in one place; they raise stress on three linked sectors: water resources, energy generation, and food production [3].

Once you put those stresses together, you get compound vulnerability. Drought, for example, can shrink water available for irrigation, reduce hydroelectric output because river flows drop, and also put direct pressure on crop performance and yields. So, to understand and handle these kinds of linked risks, you can't just zoom in on one sector and call it done. You need integrated approaches, with system-level thinking, not just sectoral optimization in isolation.

1.2 Problem Statement

Right now, Sri Lanka's agricultural system faces multiple pressures that are connected, and they need attention urgently, like really urgently. The main ones are:

- Water Resource Constraints: longer dry spells tied to El Niño reduce water supplies for crop irrigation; they also affect groundwater storage, and they lower the overall water availability for different users who compete for it.
- Energy Supply Challenges: hydroelectric plants, which support a substantial portion of national electricity, depend on sufficient water movement. When drought happens, power generation drops, so the system leans more on expensive fossil fuels.
- Agricultural Output Reduction: lower rainfall translates into reduced crop yields, and it squeezes farmer income, which becomes extra difficult for communities that rely heavily on rain-fed farming.
- Information Access Deficits: many smallholder farmers still don't get timely climate forecasts, nor agronomic guidance that is accurate enough, so decision-making happens with limited evidence.

- Monitoring System Deficiencies: the existing weather observation setup is still not strong enough, meaning it can't deliver localized information that is actionable for farming communities.

1.3 Research Gap and Justification

A lot of studies have explored climate impacts on agriculture and different adaptation tactics at the local level. Still, the literature leaves important gaps that matter for this study, and those gaps look like they remain pretty persistent:

1. Few works have looked at integrated frameworks that tackle climate forecasting, irrigation management, and crop productivity at the same time, especially in developing-country contexts [3], [9].
2. Evidence is still thin on how machine learning technologies can be adopted in smallholder farmer settings, specifically within South Asian conditions [14].
3. The adoption mechanisms and barriers for digital agricultural tools in developing country environments need deeper and more detailed investigation [14].
4. We still need a more complete analysis on how climate adaptation measures interact with water, energy food system dynamics, because the relationships aren't fully mapped [3], [9].
5. The practical implementation pathways for scaling decision support systems to national levels are not well developed yet [3].

This research aims to address each of these gaps by running a structured investigation that covers technical, social, and institutional dimensions, not only one angle.

1.4 Research Objectives and Questions

Primary Research Goal: develop, implement, and evaluate a machine learning based decision support system. The system should combine climate information, water resource management, and agricultural production optimization. The overall intention is to strengthen climate resilience for Sri Lankan farming communities facing hazards linked with El Niño.

Specific Research Questions:

1. Which neural network architectures are most effective for predicting monsoon rainfall behavior and drought conditions, three to six months ahead?
2. How can ensemble machine learning methods improve irrigation scheduling so water is conserved while crop productivity stays stable?
3. What factors have the strongest influence on farmer adoption, and on continued use of digital agricultural decision support tools?
4. When algorithm-based recommendations are applied, what measurable changes appear in water consumption, crop output, and farm income?
5. What institutional and policy arrangements can support long term operation, plus scaling of these decision support systems?

CHAPTER 2: LITERATURE REVIEW

2.1 Climate Variability and South Asian Agriculture

The El Niño-Southern Oscillation really starts from tangled interactions, between ocean temperature shifts and atmospheric pressure arrangements in the Pacific. Then those teleconnections can reach precipitation behavior, all the way thousands of kilometers away [4]. For Sri Lanka, meteorological records show that when El Niño grows stronger, the southwestern monsoon precipitation often misses climatological expectations quite a bit [2]. A bunch of regional studies point to precipitation drops that range roughly from twenty to sixty percent, depending on the monsoon stage and local geography, then it spills over into hydrology, ecology, and everyday human actions [6]. Earlier reconstructions also imply that these kinds of dry spells have been disrupting regional farming systems for centuries [7], and more recent climate forecasts still carry uncertainty, so adaptive management becomes kind of essential, because it has to handle variable conditions [8].

Too little precipitation, obviously, reduces soil moisture availability, so plants struggle even if other inputs exist. If the dryness lasts long enough, it can also mess with groundwater behavior, and then later-season water availability suffers as well [3]. In terms of farm output, drought years have been associated with reductions of about twenty to forty percent for rainfed systems,

and roughly fifteen to thirty percent for irrigated systems when water allocations get restricted [6]. That decrease then shows up as lower farmer income and, not surprisingly, higher rural poverty levels.

2.2 Water-Energy-Food System Interdependencies

The research body generally treats water, energy, and food-production systems as connected webs, not as separate “silos” that just run on their own. Water limits can reduce energy generation, especially when hydroelectric plants are forced to run at lower capacity. At the same time, energy availability steers food production because irrigation pumping depends on power. Meanwhile, what happens to food production feeds back into rural economies, and it can weaken livelihood security. So, the relationships form feedback loops, and climatic stress on water tends to create stacked vulnerabilities across the chain [3]. In extreme drought years, the pressure shows up in all three systems at once, so integrated management is more sensible than optimizing within only one sector [3], [9]. There is also work on nexus-aware interventions suggesting that improving agricultural productivity via better water management can reduce energy use for pumping too, which ends up creating extra co-benefits in multiple dimensions [9].

2.3 Machine Learning Applications in Agricultural Systems

In recent years, deep learning advances have supported forecasting tools that can detect complicated patterns inside sequential data. Some architectures are especially good for rainfall prediction because they can represent temporal dependencies and nonlinear linkages inside climate records [10], [11]. Published results indicate that accuracy can be higher by around twenty to thirty percentage points compared to older statistical approaches, and the gain seems to persist across several regions and time ranges [11], [13]. The Long Short-Term Memory (LSTM) model is used to address constraints of older recurrent networks, mainly because it can handle long-range dependencies across multiple months or seasons [11].

Ensemble techniques work by combining multiple algorithms, so the individual strengths get pooled while weaknesses are partly canceled out. These have shown value for irrigation scheduling, and Random Forest methods often perform well when dealing with nonlinear patterns, interactions among features, and missing data [12]. For monitoring crops without destruction, remote sensing helps a lot. Vegetation indices from multispectral satellite imagery often correlate strongly with biomass and crop development stage, and convolutional neural networks (CNNs) are particularly useful for pulling out spatial signals that reflect canopy growth and crop health [13].

Digital agriculture tools, if they are designed with local realities in mind, have shown a real ability to improve farming outcomes. Studies of successful deployments usually highlight simplified interfaces, offline-capable operation, and alignment with existing extension structures. Evidence from several South Asian countries indicates productivity increases in the ballpark of fifteen to thirty percent when digital support comes together with farmer training, plus active involvement from extension workers [14].

2.4 Identified Knowledge Gaps

Right now, the literature still leaves multiple holes that this study directly aims to address: there is insufficient integration of climate prediction, water optimization, and crop productivity inside unified frameworks [3], [9]; there is limited information about real-world rollout and actual adoption of AI decision tools among smallholder farmers [14]; there is not enough clarity on what enables sustained technology use in developing settings [14]; the nexus-level benefits from agricultural interventions are not analyzed deeply enough [3], [9]; and policy or institutional arrangements that could support scaling are still underdeveloped [3]. This work contributes evidence for each gap by running a systematic investigation across technical, social, and institutional layers.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design and Approach

This investigation kinda used a mixed-methods setup, meaning there was quantitative modeling and analysis paired with qualitative fieldwork, plus direct farmer engagement, so that the technical system performance is understood along with the human side, especially adoption behavior. The study sort of assumes that climate-agricultural systems have physical properties that we can measure, but at the same time, farmer decision-making is not only a math thing; it also comes from socially built knowledge systems. Overall, the work ran for 36 months, from August 2023 to August 2026, so that we could observe across multiple seasons and different climate swings.

3.2 Data Collection Strategy

Secondary Data Acquisition

We relied on existing datasets from different institutional sources. These included meteorological records, temperature, precipitation, radiation, humidity, wind- from thirty-five weather stations covering 2010 to 2025; hydrological records from reservoirs, streamflow stations, and groundwater wells; MODIS plus Sentinel-2 satellite imagery, which was processed via cloud geospatial tools to generate vegetation indices, land-surface temperature, and soil-moisture estimates on a ten-to-sixteen-day rhythm. Also included were oceanic/atmospheric El Niño indices covering 1950 to 2025 and national agricultural census data

from 2010 to 2024. After getting the secondary sources, the team did cross-checks, looked for temporal continuity gaps, flagged outliers, and compared results with what other published studies reported.

Primary Data Generation

In parallel, the team put in 200 automated weather and soil-moisture monitoring stations in the study areas starting June 2023 and kept recording through 2025. We also collected production data from 2,400 participating farmers where they documented crop type, input quantities, how much water was applied, and yields across two growing periods (2024-2025). Baseline surveys were done with 300 pilot farmers, and then there were forty semi-structured interviews, each one averaging about forty-five minutes, focused on perceptions and real experiences both mid-season and after harvest. To capture system usage, recommendation adherence, and satisfaction, we used automated logging plus post-season surveys.

3.3 Machine Learning System Components

3.3.1 Rainfall Forecasting Module

Model Type: This module used a recurrent neural network with LSTM cells, meant to catch longer temporal patterns in sequential climate data. Inputs include Pacific and Indian Ocean temperature anomalies, atmospheric circulation indices, and historical monsoon rainfall, scaled to zero mean and unit variance. Architecture includes an input layer using forty-eight timesteps across fifteen features; then three LSTM layers, each with 128 units. After each LSTM layer, we applied dropout at 0.2, and near the end, there is a fully connected output layer producing both rainfall forecasts and a confidence interval. Training followed Adam optimization (learning rate = 0.001), batch size of thirty-two, and 500 epochs, but with early stopping, where patience was twenty epochs. There was five-fold cross-validation, and the data split used 70/15/15 for train/validation/test plus a grid search for hyperparameters.

3.3.2 Irrigation Management Module

Model Architecture: an ensemble combining Random Forest and Gradient Boosting. Input variables cover current soil moisture (from sensors and satellite), weather forecasts, crop water requirements, available water supplies, soil properties, and topography. The Random Forest part was set to five hundred trees, a maximum depth of twenty, and a minimum of five leaf samples. The Gradient Boosting part used one hundred estimators with a learning rate of 0.1. Features were scaled to [0, 1]. Outputs include irrigation-volume suggestions, timing direction, and estimated water savings. Calibration was done using fifty farm-seasons of observed practice, then validated against measured changes in soil moisture.

3.3.3 Crop Productivity Estimation

Model Design: a convolutional neural network takes satellite imagery, paired with a tabular branch for other factors, and both streams merge before the later fully connected layers. Inputs include vegetation indices (NDVI, EVI, NDWI) at ten- to sixteen-day intervals, temperature and precipitation, soil properties, and historical yield records (n = 1,200 farm-seasons). In the convolutional branch, there are three layers, with 32, 64, and 128 filters, using 3x3 kernels and batch normalization. The tabular branch uses two dense layers (256 and 128 units). The final layers (128 and 64) use 0.3 dropout right before output. Yield estimates are generated about eighteen to twenty-one days before harvest. Performance is judged using R-squared, RMSE, and mean absolute percentage error

3.3.4 Farmer Recommendation Engine

This part converts the algorithmic outputs into farm-level guidance while accounting for farmer conditions and what they prefer. Delivery is through a mobile application, SMS alerts for urgent time-based decisions, community radio announcements, and short briefings from extension workers. Recommendations are adjusted based on what resources are realistically available, which crop varieties are accessible, water-access constraints, and the farmer risk tolerance level.

3.4 Geographic and Agroecological Context

This study was carried out across five separate agroecological zones, capturing the main climate and production rhythms, and each zone had sixty pilot farms, so the overall spread stayed balanced. The settings were classified as follows,

Zone	Rainfall (mm/yr)	Elevation (m)	Key Crops	Pilot Farms
Arid	600-900	0-300	Rice, cereals, pulses	60
Semi-Humid	1,250-1,750	300-600	Rice, specialty/tree crops	60
Humid	2,250-3,500	0-300	Tea, tree crops, vegetables	60
Highland	1,800-2,500	600-1,000	Tea, temperate crops	60
Coastal	1,200-2,000	0-100	Tree crops, aquaculture	60

3.5 Analysis Methods

Quantitative Analytical Approach

Quantitative Analytical Approach

To judge model performance, we used mean absolute error, root mean squared error, correlation coefficients, plus categorical accuracy specifically for drought prediction. The impact portion compared results between system users and control farmers, using independent-samples t-tests, chi-squared tests, and multivariate regression modeling. For the temporal side, autoregressive integrated moving-average models were applied to inspect seasonal cycles and the small-to-large shifts across years. At the farm scale, outcomes included water consumption, crop output, farm revenues, and energy expenses.

Qualitative Investigation Approach

In parallel, forty farmer interviews were processed through qualitative data-management software, and coding was handled in an iterative way. Theme building followed constant comparison, spanning farmer group differences and the agroecological zones themselves. Field collection continued until interviews stopped bringing up genuinely new themes, then the qualitative insights were cross-checked against the quantitative results, so interpretation stayed not only broad, but also more anchored.

3.6 Ethical Framework

The protocol was reviewed and granted approval by the university research ethics committee before the actual implementation began (approval number UPER-2023-048). Farmers signed written informed consent after being informed about the study aims, procedures, how data would be used, and what their participation rights were, in the language they preferred. For data handling, anonymity was maintained, with reporting done in aggregated form and identification codes used for every farmer record. During the pilot phase, participating farmers obtained tool access without charge, results were shared through extension networks, and all information was stored on secure, access-controlled servers, with no personally identifying information included in any publications.

CHAPTER 4: RESULTS AND FINDINGS

4.1 Predictive Model Performance

4.1.1 Climate Forecasting Outcomes

The system correctly predicted rainfall direction in roughly sixty-eight to seventy-two percent of the three-month forecasts, and it flagged drought episodes in about seventy-five percent of the predictions. The false alarm rate sat around twelve percent. When the horizon stretched to six months, accuracy went down to fifty-eight to sixty-five percent, which somewhat matches the well-known climate predictability limits at longer lead times, but still, it stays well above baseline-style methods. If you look at persistence-based forecasting, it only managed thirty-five to forty-two percent accuracy, meaning the developed system improved things by something like twenty-six to thirty-seven percentage points. In the validation span, it nailed eight out of ten big drought episodes, and the skill was best at the three-month lead window, which lines up with farmers' crop variety choices and field preparation schedules.

4.1.2 Irrigation Recommendation Performance

The irrigation suggestions, as implemented, reduced irrigation volume by an average of 27.3% (but it ranged from twenty-three to thirty-five percent when you break it down by zone and crop). Water savings were strongest in the drier places, and then weaker in the wetter areas, which suggests the calibration stayed sensible with local conditions. Also, the results show water reduction happened alongside yield maintenance or even a bit of yield lift. It looks like real efficiency gains rather than any production compromise. Predicted irrigation volume was off from the optimal by about eight to twelve millimeters on average, and the prediction intervals managed to cover about ninety percent of the observed optimal values.

4.1.3 Crop Productivity Forecasting

For yield, the models showed a strong statistical fit. For rice, $R^2 = 0.82 \pm 0.06$ with $n = 120$ farm seasons, and for vegetables $R^2 = 0.76 \pm 0.08$ with $n = 80$ farm seasons. The forecasts came out eighteen to twenty-one days before harvest, giving enough time for harvest logistics and also market engagement, while the confidence intervals communicated how uncertain the predictions were.

4.2 Field Pilot Implementation Outcomes

These outcomes were matched between three hundred system users and three hundred control farmers, over two growing seasons.

Water Resource Utilization

Water Use Metric	Control	System Users	Change	Sig.
Seasonal application (mm)	650 ± 85	485 ± 78	-25.4%	p<0.001
Water productivity (kg/mm)	6.5 ± 1.2	8.2 ± 1.1	+26.2%	p<0.001
Irrigation applications (no.)	8.2 ± 2.1	5.4 ± 1.8	-34.1%	p<0.001

Users lowered irrigation frequency while simultaneously applying more water per event, which points to better timing, tied to soil moisture information. And the higher productivity per unit water is basically consistent with better water delivery to actual crop demand.

Crop Production and Energy Outcomes

Crop	Control (kg/ha)	User (kg/ha)	Change	p-value
Rice	4,200 ± 420	5,050 ± 380	+20.2%	<0.001
Vegetables	18,500 ± 2,200	21,800 ± 1,900	+17.8%	<0.001
Maize	3,800 ± 450	4,320 ± 420	+13.7%	0.01
Legumes	1,500 ± 200	1,680 ± 180	+12.0%	0.03

All these improvements stayed statistically significant, and they repeated across both seasons, plus across every geographic zone. Meanwhile, irrigation pumping energy dropped a lot for system users. It went from 980 ± 180 to 720 ± 150 kWh/ha per season (-26.5%), and pumping cost moved from LKR 55,500 ± 12,600 down to LKR 38,100 ± 11,400 per hectare (-31.4%). That translated into an average farm saving of LKR 17,400 per season, which is around seven to ten percent of the full farming expenses.

Farm Economic Performance

Economic Indicator	Control (LKR)	System Users (LKR)	Change	p-value
Seasonal gross income	555,000	687,000	+23.8%	<0.001
Net income after expenses	375,000	504,000	+34.4%	<0.001

Note: monetary values are reported in Sri Lankan Rupees (LKR), converted from the study's original USD figures at an approximate rate of LKR 300 = USD 1 for the 2023–2025 study period.

The income improvements held up across back-to-back seasons and across geographic regions. In plain terms, it means a meaningful jump for rural livelihood and food security.

Drought Resilience During the 2023-2024 El Niño Event

Performance During Drought	Control	System Users	Advantage
Yield loss	-38.5%	-10.8%	27.7 pts
Water conservation	—	41.2%	—
Income loss	-42.3%	-14.6%	27.7 pts

System users kept about eighty-eight percent of normal production during drought, while control farms held only around sixty-one percent. That gap of roughly twenty-seven percentage points supports strong livelihood protection under climate pressure.

4.3 Farmer Perspectives and Experiences

System Usability and Barriers

About seventy-eight percent of farmers said the application interface felt workable after one to two training sessions, and the text alerts were especially liked for time-sensitive decisions. Also, radio broadcasts seemed to actually reach farmers who had limited smartphone access. But the remaining twenty-two percent asked for interface streamlining, while offline functionality was the big deal in zones where connectivity is a bit messy.

When it came to “why not”, adoption hurdles showed up like resource unavailability at thirty-one percent, digital-skill gaps at twenty-seven percent, and water-access constraints at twenty-four percent. There were also trust-confidence frictions right at the start, reported by nineteen percent, and connectivity limitations at fifteen percent.

Adoption Support Factors

In terms of what helped, extension-worker endorsement was the strongest driver (eighty-eight percent among those who adopted). Farmers also pointed to visible production improvements in the first season, reported by ninety-two percent. Peer networks mattered too, with seventy-six percent, and system customization to fit local conditions came in at seventy-one percent. Financial supports, credit or input-subsidy linkages, were cited by sixty-four percent as an important nudge.

Farmer-Perceived Impacts and Suggestions

Farmers described that precision irrigation scheduling cut water use, while still keeping production steady. And in the drought year, people basically saw the value quickly; participants maintained realistic output even when neighbors experienced serious losses.

Still, there were limits they perceived. Some recommendations seemed to presume water availability that irrigation authority allocations did not always guarantee. Input-purchase suggestions were also flagged because in some cases they overshot what farmers could afford.

For improvements, farmers suggested market-price integration (forty-two percent), input-availability information (thirty-four percent), more local language interfaces (twenty-eight percent), full offline operation (nineteen percent), plus deeper incorporation of traditional knowledge (fifteen percent).

4.4 Adoption and Sustained Usage Patterns

Across three hundred pilot farmers over two seasons, mobile application logins averaged 23.4 ± 9.2 per season. The text-alert opening rate sat at eighty-one percent, and sixty-eight percent of farmers said they took action on recommendations. In broadcast areas, seventy-three percent listened to the radio programming, and the overall recommendation implementation rate was sixty-four percent.

Implementation moved upward from fifty-two percent in the first season to sixty-four percent in the second, which suggests that people learned more and became more confident over time.

Characteristic	Early Adopters (n=45)	Later Adopters (n=255)	Non-Adopters (n=30)
Average farm size (ha)	1.2	0.8	0.6
Smartphone ownership	96%	71%	22%
Extension contacts/season	8.4	4.2	1.1
Prior technology adoption	78%	41%	8%
Formal education (years)	11.2	8.6	5.4

Large farms, higher digital literacy, steady extension engagement, and earlier technology exposure were strongly associated with higher adoption likelihood.

CHAPTER 5: DISCUSSION AND CONCLUSIONS

5.1 Findings Summary and Interpretation

Technical System Viability

Overall, this investigation shows that machine learning-based decision-support systems do work, pretty well, for climate adaptation in tropical developing-country agriculture. The three-month-ahead forecasts were right in roughly seventy percent of cases, and that beats the baseline forecasts in a fairly clear way. What stands out is that the system is especially strong at spotting drought, and those drought flags line up with what farmers actually need in practice. On the irrigation side, optimization delivered about a twenty-seven percent reduction in water use, while crop productivity stayed the same or even improved. So, it is like a double win: less water stress, but not at the expense of yields. Crop output gains, in the range of twelve to twenty percent, show up across multiple crops and seasons. There is also decent consistency across different geographic areas, which suggests the approach is robust even when environmental conditions shift. But the most meaningful part, perhaps, is what happened during real drought conditions: farmers using the system reached about eighty-eight percent of normal production, compared with sixty-one percent for non-users. That creates a twenty-seven-percentage-point gap which, in livelihood terms, really looks like substantial protection.

Socioeconomic and Adoption Findings

Net farm income went up by thirty-four percent for system users. The increase seems tied to combined pathways, higher yields, lower input costs, and better operational efficiency. The results also hint at a broader “nexus” effect: water demand drops, so pressure on limited resources is reduced. At the same time, irrigation energy demand drops too, and this can free hydropower for broader use, while more production supports food security and rural employment. Even so, adoption is not perfectly even. Adoption rates were lower among the smallest farms. This is interesting, because those farms might have had larger benefit-to-cost ratios. That suggests that when scaling, targeted support for marginal farmers could be important, not optional. Extension-worker validation, visible benefits, peer discussion, and local tailoring were the strongest adoption drivers. Looking ahead, sustained engagement will probably need dependable system performance, ongoing technical assistance, routine seasonal updates, and continued integration with extension services, not just a one-time introduction.

5.2 Implications for Agricultural Practice

Farming Operations

- Shift from calendar-based irrigation scheduling to condition-based scheduling, using soil moisture info along with weather forecasts.
- Apply climate forecasts to pick crop varieties earlier, in ways that match expected precipitation patterns.
- Bring crop diversity into practice to reduce single-point risk, and keep records that track decisions as well as outcomes.

- Involve extension workers to interpret recommendations and translate them to the local situation.

Extension Service Integration

- Treat decision-support systems as complements to extension, not as something that replaces extension expertise.
- Strengthen extension training focused on reading climate information and adapting recommendations on-farm.
- Use farmer groups plus lead-farmer models to spread recommendations efficiently, while building feedback loops so the system improves over time.

5.3 Policy and Institutional Implications

Government Priorities

- Set up institutional structures and secure funding so national-level system operation and maintenance is realistic.
- Put money into weather station and satellite monitoring networks, since those generate the core forecasting data.
- Support workforce training programs for climate-smart agriculture and digital tool use.
- Adjust irrigation allocation policy so it can align with climate forecasts, enabling proactive drought response.
- Coordinate across agriculture, water, and energy sectors, while also creating data security and privacy standards.

Development Partner Support

- Build institutional capacity for long-term, durable system operations.
- Fund research that tackles implementation barriers, while also exploring optimization opportunities.
- Help with technology transfer and knowledge exchange, especially with neighboring nations facing comparable constraints.

5.4 Research Limitations

A few scope limits should be kept in mind while reading the findings: the two-year pilot covered only two agricultural seasons, so a longer observation period would likely solidify conclusions about sustained effects. The five agroecological zones cover major regions, but they do not fully capture every microclimatic nuance. Only forty farmer interviews were used; these are useful and illustrative, yet not statistically generalizable. Also, limited demographic coverage on age, gender, and wealth narrows the ability to analyze equity outcomes thoroughly. Finally, the work documented productivity and resource-use impacts, but it did not include a full cost analysis for national-scale operation and maintenance.

Unresolved Research Questions - Still open: what happens to long-term adoption when subsidized support ends, and real operational costs at scale become the norm. Another question is how model performance and forecast predictability might change under future climate regimes. There is also the matter of realistic national deployment economics, including financial sustainability for smallholders. Social networks and trust in institutions are likely relevant too, but they were not fully addressed. Lastly, it remains unclear how decision-support tools fit together with complementary adaptation measures such as crop insurance and water harvesting.

5.5 Future Research Directions

- Do extended longitudinal work; three to five years would be helpful, following farmer cohorts, skill development, and livelihood trajectories.
- Study scaling and financing, including institutional models, revenue mechanisms, and policy frameworks needed for national implementation.
- Run gender-disaggregated equity analysis to see whether adoption and benefits differ across demographic groups.
- Carry out climate-resilience assessments, testing system effectiveness under projected climate scenarios using downscaled model outputs.
- Investigate sectoral integration, linking agricultural decision support with energy and water-sector planning.
- Conduct regional adaptation studies, testing whether the framework transfers well to neighboring nations with similar baseline conditions.

5.6 Conclusions

The research presented here demonstrates that machine-learning-based decision-support systems can be used in Sri Lanka to enhance climate-resilient agriculture and address water, energy, and food security challenges.

Core contributions include (1) the development and validation of one integrated machine-learning framework that kind of ties together climate forecasting, irrigation optimization, and crop monitoring, specifically for a tropical developing-country context; (2) quantified proof of how well the system works, including water use down (-27%), productivity up (+13 to 20%), income improved (+34%), and drought resilience stronger (with a 27-percentage-point jump in yield maintenance). Also (3) we identified the adoption-enabling factors, like extension-worker support, visible results, farmer networks, and local customization. Then (4) it's shown how agricultural optimization can simultaneously help with water conservation, energy efficiency, and food production, not one thing only. Finally (5), there is operational validation via field testing over three hundred farms during two seasons, and it highlights the practical requirements needed for implementation to get it running day by day.

As climate variability intensifies and El Niño events threaten agricultural systems across South Asia, decision-support technologies incorporating artificial intelligence and climate forecasting are essential tools for building adaptive capacity. There is evidence that such systems, when appropriately designed and deployed with institutional support, substantially enhance both farmer resilience and production sustainability. Scaling digital agriculture requires government commitment to infrastructure, sustained funding for system operation, integration with extension services, farmer-centered design, and regional knowledge sharing. Sri Lanka can become a regional leader in climate-smart agriculture by investing in decision-support systems and supporting institutional reforms - benefits that include improved food security, water sustainability, farmer resilience, and economic opportunity justify the necessary commitments.

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APPENDICES

Appendix A: Workshop Integration

Methodology was informed by four training workshops covering research-question formulation and literature positioning; critical evaluation of study quality and methodology; the five-chapter research structure with primary/secondary data integration; and applied use of AI tools (ML platforms, geospatial analysis, cloud computing, LLMs) for research acceleration.

Appendix B: Data Quality Procedures

Secondary data underwent temporal-gap interpolation, range checks, cross-source validation, and metadata documentation. Primary data quality was ensured through annual sensor calibration, cross-checking of agricultural records against field measurements, interview-transcription verification, and independent qualitative-coding reliability checks.